

Point Cloud Upsampling on ModelNet40

Geometry Consistency with Original Point Clouds and PointNet++ Classification

- Geometry reference: corresponding Original point cloud only (no mesh / no P2F)
- Unequal cardinality (Line A 4096 vs 1024; Line B Downsampled 256 vs 1024) = discrete-set consistency only
- PointNet++ retrained per input size (A: 4096 from scratch; B: 256 / 1024 from scratch)
- Two-line protocol with EAR, PDANS, PU-Net, PU-GCN, PU-EdgeFormer
- Companion editable PPTX has 21 slides

Geometry Evaluation Definition

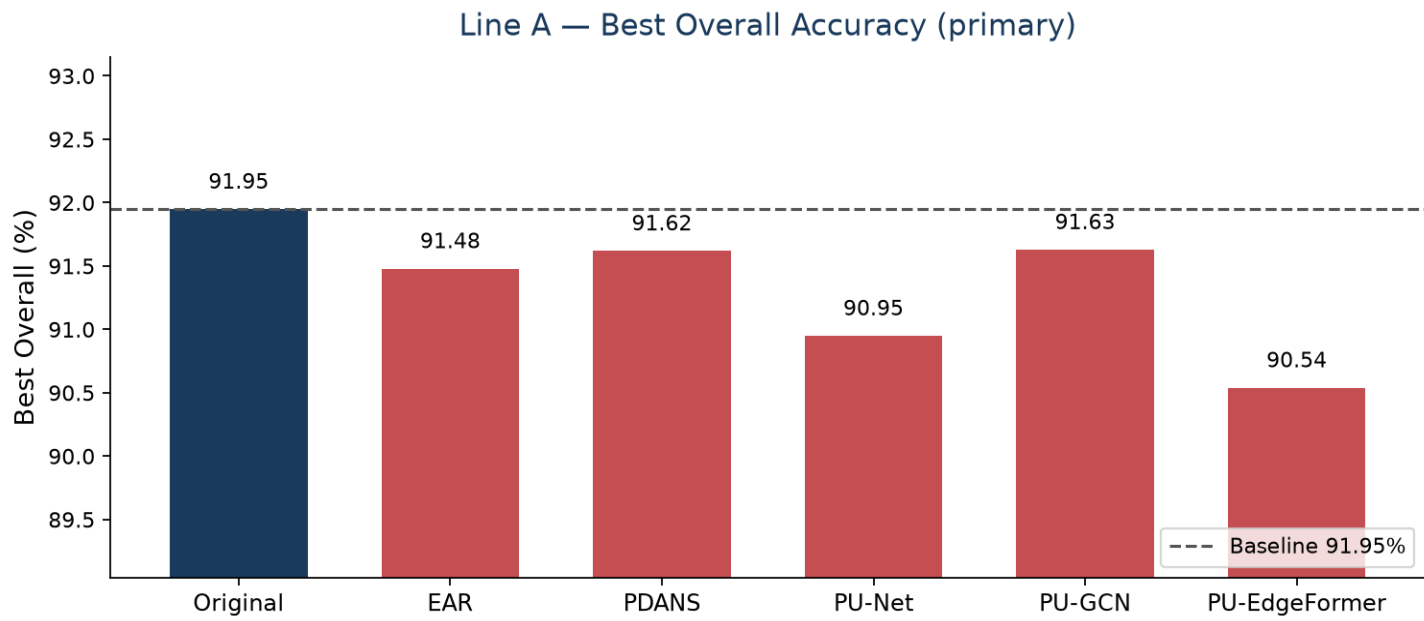
- Equal cardinality only. Mesh-ref 256/4096: area-weighted from ModelNet40 .off + unit sphere.
- Line A: 4096 vs mesh-ref 4096. Line B Downsampled: 256 vs mesh-ref 256. Line B ups: 1024 vs Original 1024.
- Self rows: CD=0, HD=0. Unequal-|P| CD/HD omitted.

Geometry Results (equal cardinality)

Line	Method	Points	Reference	CD	HD	NUC	Δ NUC	N
A	Mesh-ref 4096	4096	Mesh-ref 4096	0.0000	0.0000	0.6419	0.0000	2468
A	EAR	4096	Mesh-ref 4096	0.0547	0.1154	0.8156	0.1738	2468
A	PDANS	4096	Mesh-ref 4096	0.0468	0.1135	0.6612	0.0193	2468
A	PU-Net	4096	Mesh-ref 4096	0.0497	0.1112	0.6052	-0.0366	2468
A	PU-GCN	4096	Mesh-ref 4096	0.0423	0.0932	0.5204	-0.1215	2468
A	PU-EdgeFormer	4096	Mesh-ref 4096	0.0467	0.1109	0.6754	0.0336	2468
B	Mesh-ref 256	256	Mesh-ref 256	0.0000	0.0000	2.1265	0.0000	2468
B	Downsampled $\times 4$	256	Mesh-ref 256	0.1333	0.2049	2.0933	-0.0332	2468
B	Original	1024	Original 1024	0.0000	0.0000	1.0836	0.0000	2468
B	EAR	1024	Original 1024	0.0767	0.1932	0.9653	-0.1183	2468
B	PDANS	1024	Original 1024	0.0627	0.1905	1.0528	-0.0307	2468
B	PU-Net	1024	Original 1024	0.0646	0.1083	1.0801	-0.0035	2468
B	PU-GCN	1024	Original 1024	0.0558	0.1514	1.5240	0.4404	2468
B	PU-EdgeFormer	1024	Original 1024	0.0619	0.1776	1.1225	0.0390	2468

Line A Classification (Best Overall primary)

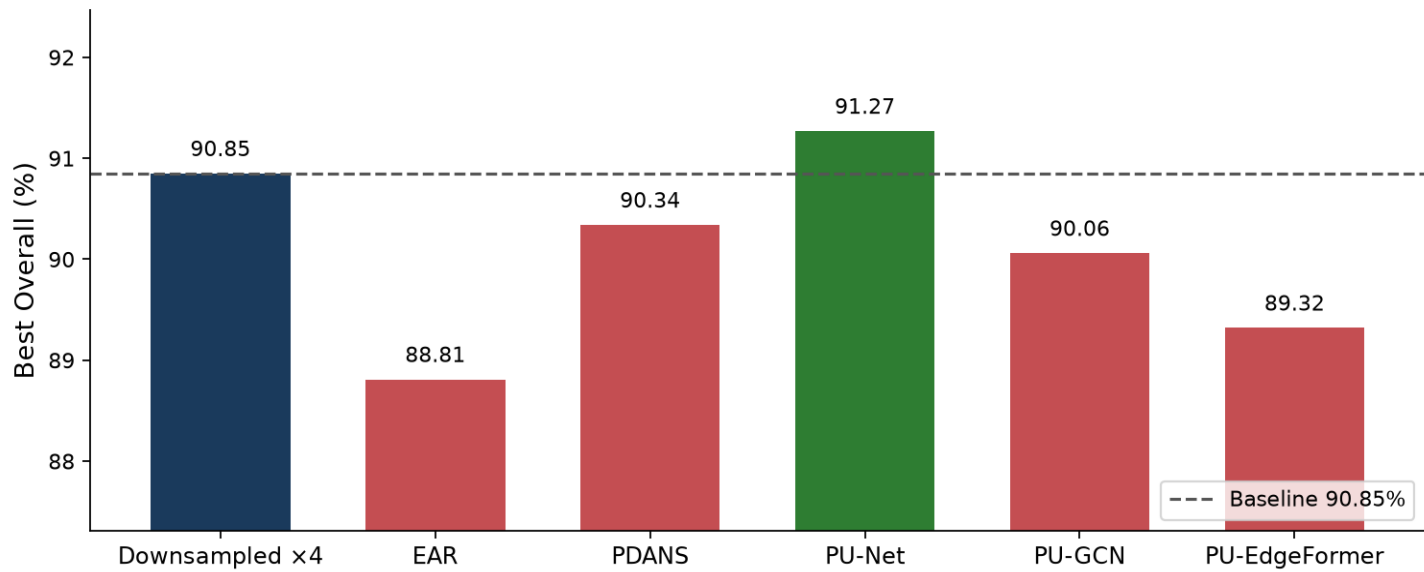
Method	Pts	Best Overall	Final Overall	Δ Best vs Original
Original	1024	91.95%	91.31%	+0.00 pp
EAR	4096	91.48%	90.85%	-0.47 pp
PDANS	4096	91.62%	90.87%	-0.33 pp
PU-Net	4096	90.95%	90.55%	-1.00 pp
PU-GCN	4096	91.63%	91.00%	-0.32 pp
PU-EdgeFormer	4096	90.54%	90.13%	-1.41 pp



Line B Classification (Best Overall primary)

Method	Pts	Best	Final	Δ vs Down	Gap vs Original
Downsampled $\times 4$	256	90.85%	90.46%	+0.00 pp	-1.10 pp
EAR	1024	88.81%	88.02%	-2.04 pp	-3.14 pp
PDANS	1024	90.34%	89.46%	-0.51 pp	-1.61 pp
PU-Net	1024	91.27%	90.65%	+0.42 pp	-0.68 pp
PU-GCN	1024	90.06%	89.63%	-0.79 pp	-1.89 pp
PU-EdgeFormer	1024	89.32%	88.58%	-1.53 pp	-2.63 pp

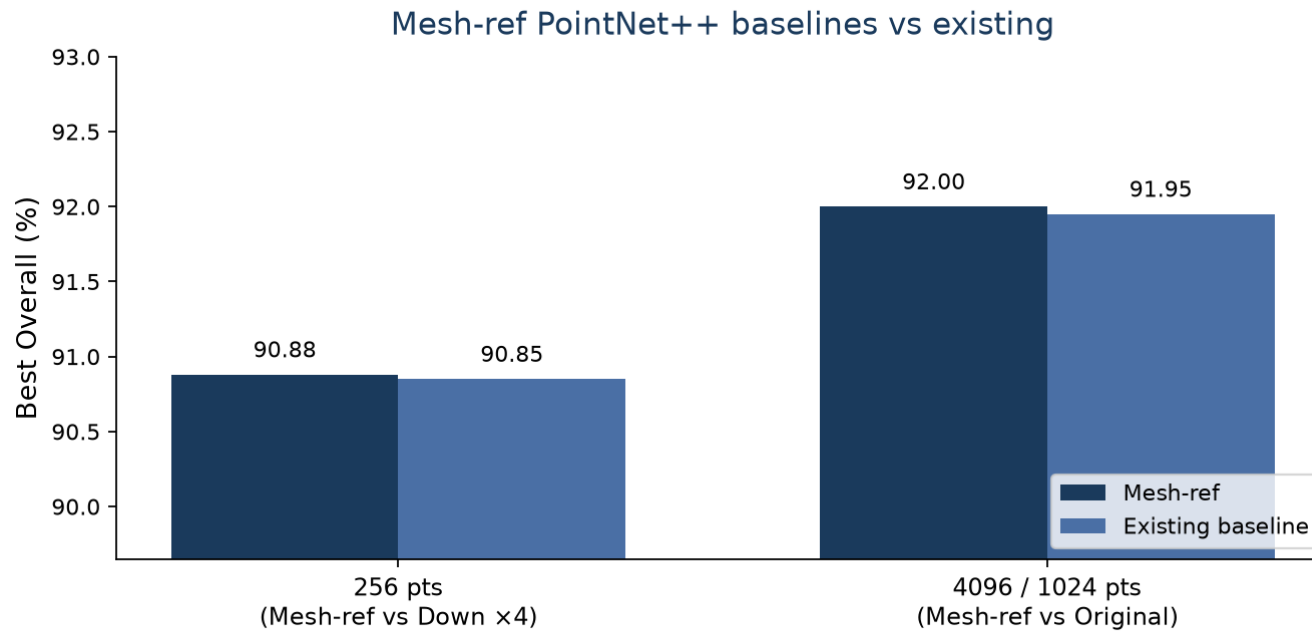
Line B — Best Overall Accuracy (primary)



Mesh-ref PointNet++ Baselines

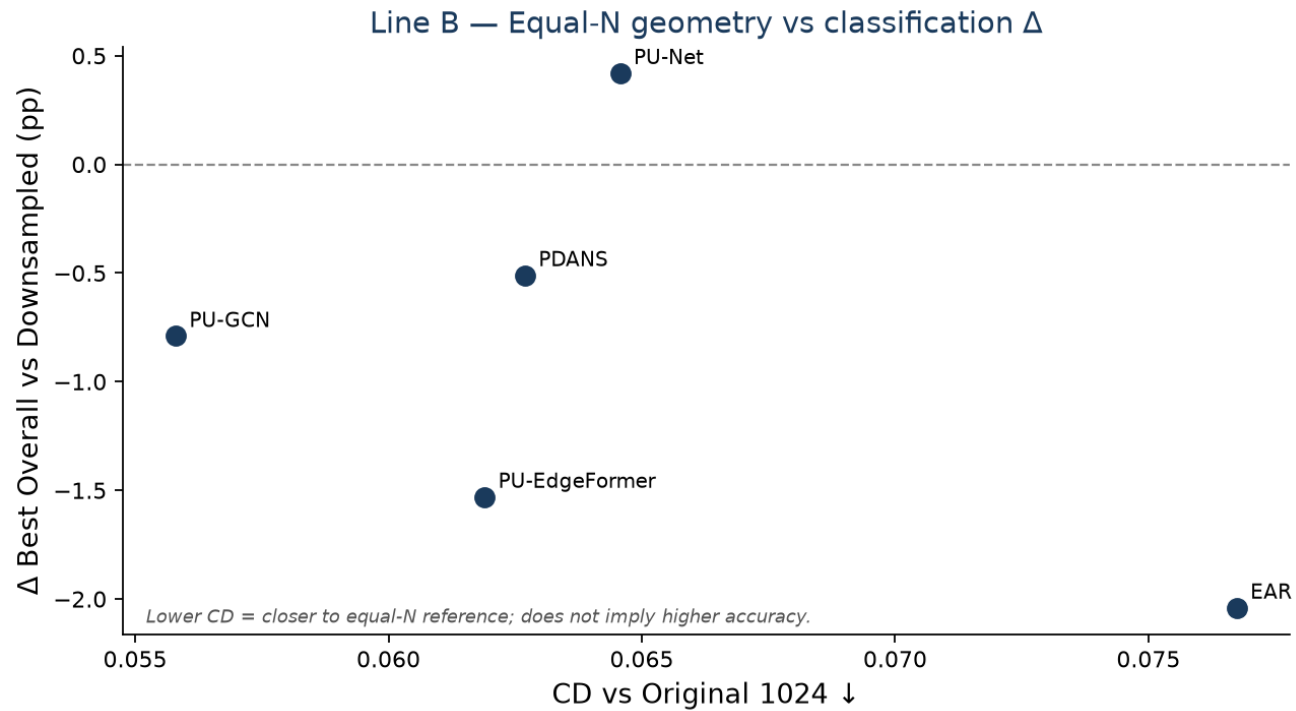
Method	Pts	Best Overall	Final Overall	Compare to	Δ Best
Mesh-ref 256	256	90.88%	90.38%	Downsampled $\times 4$ 256 (90.85%)	+0.03 pp
Mesh-ref 4096	4096	92.00%	91.18%	Original 1024 (91.95%)	+0.05 pp

- Independent area-weighted samples from ModelNet40 .off (not downsampled from Original 1024).
- Mesh-ref 256 $\approx$ Downsampled $\times 4$ 256 (90.88% vs 90.85%, +0.03 pp).
- Mesh-ref 4096 $\approx$ Original 1024 (92.00% vs 91.95%, +0.05 pp) — not a meaningful margin.



Geometry vs Classification

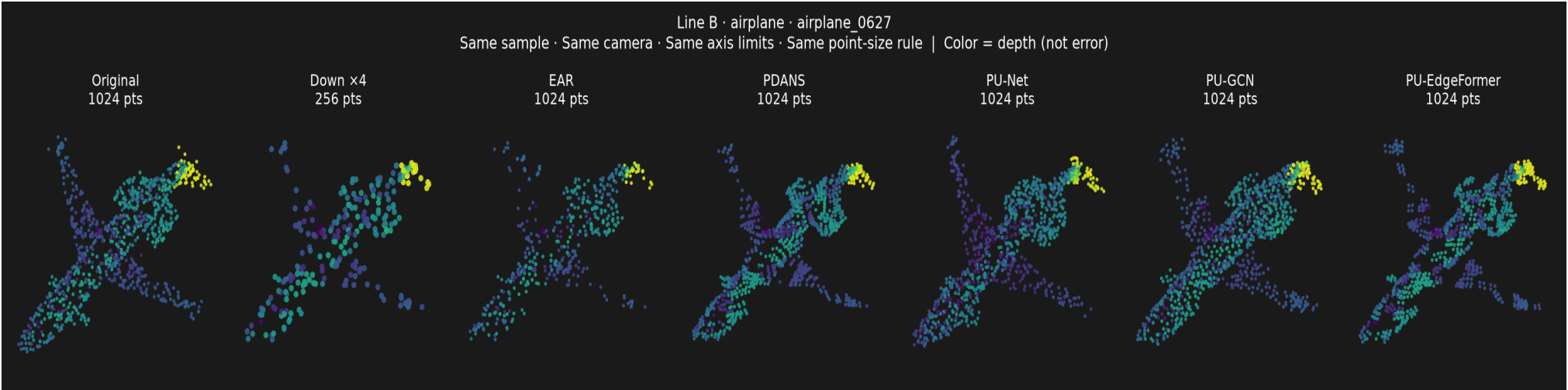
- Lower CD/HD = greater consistency with discrete Original; does not automatically imply higher accuracy.
- Limited methods: no strong statistical claim; geometry alone is insufficient to predict classification.



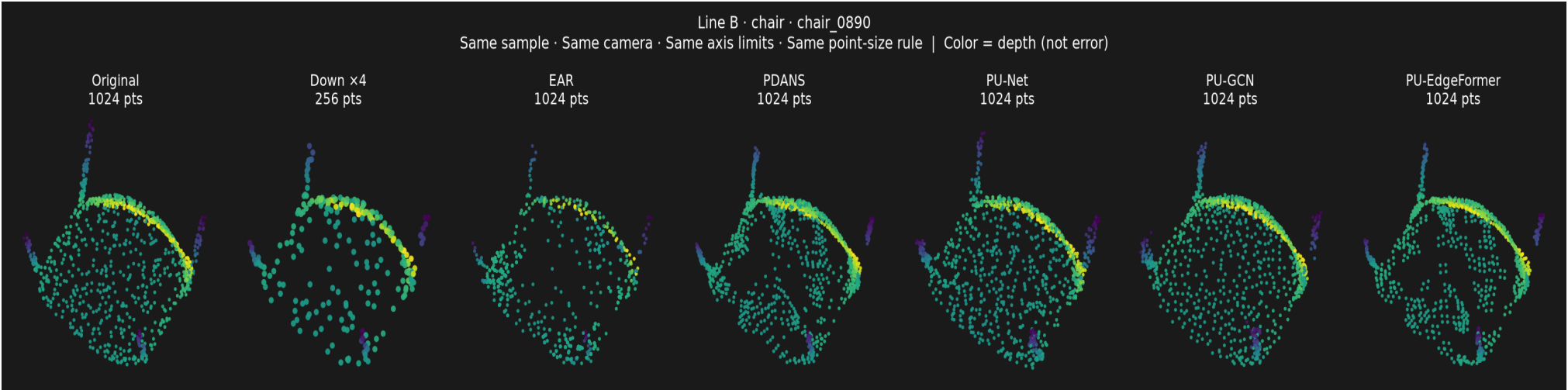
Main Findings / Conclusions

- Line A: Original 1024 remains best at 91.95% among densification branches; 4096 upsamplers do not improve OA.
- Line B: PU-Net only above Downsampled 256 (91.27%, +0.42 pp); still -0.68 pp below Original.
- Mesh-ref 256 $\approx$ Downsampled 256 (90.88%, +0.03 pp); Mesh-ref 4096 $\approx$ Original 1024 (92.00%, +0.05 pp).
- Denser mesh sampling (4096) does not beat Original 1024 by a meaningful margin.
- Geometry CD/HD: equal-N mesh-ref 256/4096 + Original 1024 for Line B recovery.
- Upsampling quality is method-dependent and task-dependent.

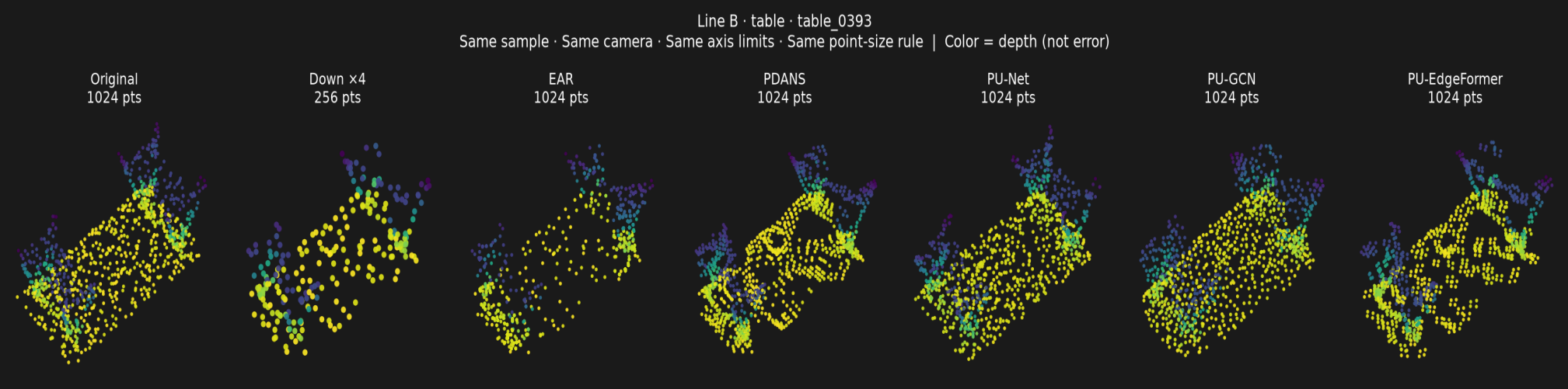
Line B qualitative — airplane



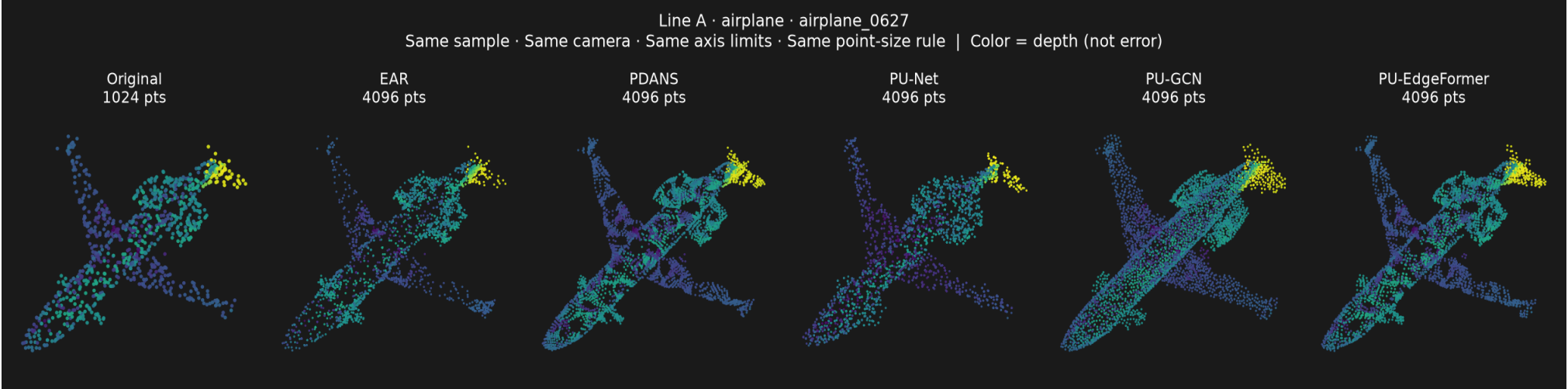
Line B qualitative — chair



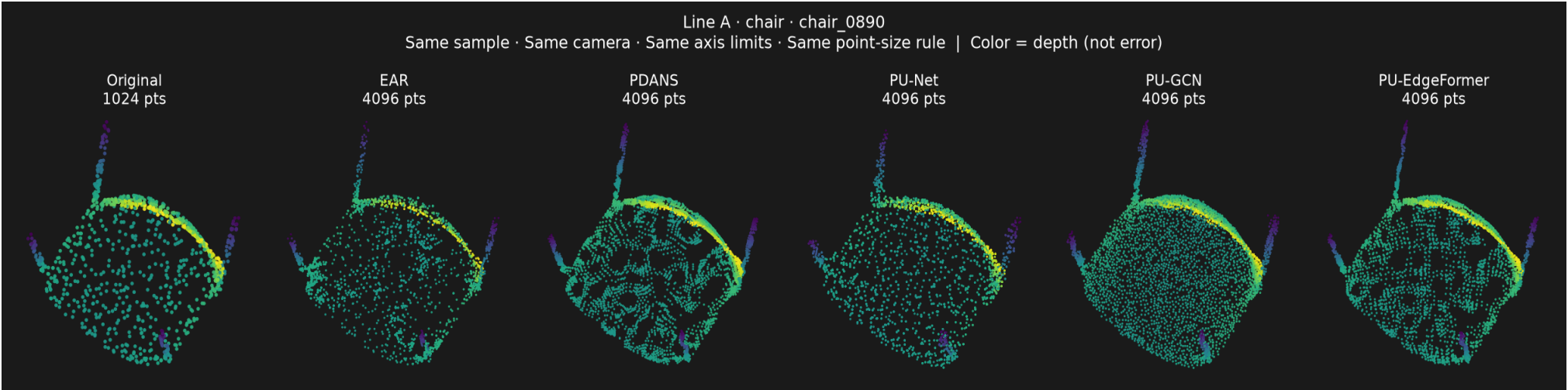
Line B qualitative — table



Line A qualitative — airplane



Line A qualitative — chair



Line A qualitative — table

